

29-3c The Money Multiplier

The creation of money does not stop with First National Bank. Suppose the borrower from First National uses the **\$90** to buy something from someone who then deposits the currency in Second National Bank. Here is the T-account for Second National Bank:

Second National Bank			
Assets		Liabilities	
Reserves	\$ 9.00	Deposits	\$90.00
Loans	81.00		

After the deposit, Second National has liabilities of **\$90**. If Second National also has a reserve ratio of **10** percent, it keeps assets of **\$9** in reserve and makes **\$81** in loans. In this way, Second National creates an additional **\$81** of money. If this **\$81** is eventually deposited in Third National Bank, which also has a reserve ratio of **10** percent, Third National keeps **\$8.10** in reserve and makes **\$72.90** in loans. Here is the T-account for Third National Bank:

Third National Bank			
Assets		Liabilities	
Reserves	\$ 8.10	Deposits	\$81.00
Loans	72.90		

The process goes on and on. Each time that money is deposited and a bank loan is made, more money is created.

How much money is eventually created in this economy? Let’s add it up:

Original deposit	= \$100.00
First National lending	= \$ 90.00 (= .9 × \$100.00)
Second National lending	= \$ 81.00 (= .9 × \$90.00)
Third National lending	= \$ 72.90 (= .9 × \$81.00)
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•	•
•	•
Total money supply	= \$1,000.00

It turns out that even though this process of money creation can continue forever, it does not create an infinite amount of money. If you laboriously add the infinite sequence of numbers in the preceding example, you find that the **\$100** of reserves generates **\$1,000** of money. The amount of money the banking system generates with each dollar of reserves is called the

money multiplier (the amount of money the banking system generates with each dollar of reserves)

. In this imaginary economy, where the **\$100** of reserves generates **\$1,000** of money, the money multiplier is **10**.

What determines the size of the money multiplier? It turns out that the answer is simple: *The money multiplier is the reciprocal of the reserve ratio*. If R is the reserve ratio for all banks in the economy, then each dollar of reserves generates $1/R$ dollars of money. In our example, $R = 1/10$, so the money multiplier is **10**.

This reciprocal formula for the money multiplier makes sense. If a bank holds **\$1,000** in deposits, then a reserve ratio of **1/10** (**10** percent) means that the bank must hold **\$100** in reserves. The money multiplier just turns this idea around: If the banking system as a whole holds a total of **\$100** in reserves, it can have only **\$1,000** in deposits. In other words, if R is the ratio of reserves to deposits at each bank (that is, the reserve ratio), then the ratio of deposits to reserves in the banking system (that is, the money multiplier) must be $1/R$.

This formula shows how the amount of money banks create depends on the reserve ratio. If the reserve ratio were only **1/20** (**5** percent), then the banking system would have **20** times as much in deposits as in reserves, implying a money multiplier of **20**. Each dollar of reserves would generate **\$20** of money. Similarly, if the reserve ratio were **1/4** (**25** percent), deposits would be **4** times reserves, the money multiplier would be **4**, and each dollar of reserves would generate **\$4** of money. Thus, *the higher the reserve ratio, the less of each deposit banks loan out, and the smaller the money multiplier*. In the special case of **100-percent-reserve banking**, the reserve ratio is **1**, the money multiplier is **1**, and banks do not make loans or create money.